Re-engineering the Student Renewal -> Class Scheduling Pipeline

A zero-touch, AI-assisted operating model with a working reconciliation prototype

Role **AI Program Manager (Operations) - Stage 2 take-home**
Company **Bambinos.live**
Prepared for **Head of Operations**
Candidate **Virender Gupta**
Evidence base **Official sample: 13 invoices, 12 faculty forms, 4 booking requests, 2 schedule rows, 13 reference students (snapshot 2026-11-03 18:00)**

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1. Executive summary

Bottom line. In the sample, 11 of 13 invoices are paid but only 1 has a scheduled class, and 5 of the 11 have waited more than 24 hours. The delay is not teacher speed (median 27 minutes when a form is submitted); it is missing forms, manual approval of clean renewals, and approvals that later cannot be booked. A deterministic engine can safely clear the clean cases and give every other case a named owner.

- **What the data shows.** 3 paid invoices have no exact-id teacher form (84906, 84910, 84913); form 3313 cites a non-existent invoice (84931, a digit transposition of 84913); 2 of the 4 forms Ops approved (3305, 3312) could not be booked; clean invoice 84903 has waited about 33 hours for approval; booking 5502 is validated yet unscheduled for 49 hours.
- **What I built.** A Python reconciliation engine (16 checks plus a decision step, audit trail, idempotency ledger), exception ownership, follow-up drafts, a Streamlit dashboard, SOP, 30-60-90 plan and risk register. Automated tests: **44 passed** (run when this PDF was built).
- **What it decides on the sample.** 2 auto-approved (84903, 84908), 1 ready to schedule on an existing booking (84902), 1 already processed (84901), **9 exceptions plus 1 orphan form**. Re-running changes nothing: 0 new executions, 3 skipped.
- **Process insight.** For same-batch renewals the teacher form adds nothing the invoice and reference table lack. A *simulation* that makes it optional lifts clean auto-approvals from 2 to 5 of 13; the form stays mandatory for batch changes.
- **Value (assumption-based).** About 26 to 64 hours of manual checking avoided per 1,000 renewals a month, if each avoided check takes 10 minutes. Volume and minutes are assumptions to validate; the stronger case is latency.
- **Decisions needed from you:** (1) does the batch seat count already include the renewing student? (2) may classes start after instalment 1? (3) who approves discounts? (4) may the form become optional for same-batch renewals? (5) names for the six proposed owner roles.

Honest scope. Built and run: engine, tests, drafts, dashboard layout. **Not built or connected:** n8n (the workflow imports and runs locally in n8n 2.35.7 with a manual trigger, but its schedule trigger never fired and its notification nodes are placeholders), Razorpay/Stripe webhooks, WhatsApp Business API, production database, live LLM calls. The sample is small, so percentages illustrate method and are not a production baseline.

2. Business context and the existing process

Bambinos.live runs live small-group classes for children aged 4-16 across India, GCC and US/UK. Every renewal passes four manual hand-offs. The table pairs each stage with what the sample actually shows.

Stage	Today (manual)	Observed failure in the sample
1 Invoice	Parent pays via Razorpay/Stripe link	1 Failed (84911), 1 Partially_Paid (84909), 1 discounted without approver (84907), 1 payer differs from registered contact (84904)
2 Faculty form	Teacher keys invoice link, amount, preferred slot	3 paid invoices without form; typo'd invoice id (3313 -> 84931); duplicate (3308/3309); form for an unpaid invoice (3311); amount overstated (3310)
3 Booking request	Ops verifies payment, approves, booking created	2 of 4 approved forms unbookable (3305 batch 6/6, 3312 batch change with NULL batch); 8 of 12 forms still pending; clean 3303 pending ~33 h
4 Class schedule	Parent/Ops picks batch; sessions scheduled	3 of 4 bookings have no schedule rows; 5502 waiting 49 h; 5501 has 2 schedule rows for 24 purchased sessions

Why this matters. A paid renewal that does not become a scheduled class is a customer experience failure and a revenue-recognition risk. Each stage also relies on a person remembering to act, with no timer and no owner.

3. Data understanding and methodology

All five official CSVs were downloaded from the assessment repository into `data/raw/` and are never modified (SHA-256 verified before and after every run; tested). Missing values are the literal string NULL and are mapped to true nulls. Soft-deleted rows would be logged, never silently dropped.

File	Rows	Key / link	Notes
1_invoices	13	id; student_id, user_id	11 Paid, 1 Partially_Paid, 1 Failed; all currency INR (also Stripe +971 / +1)
2_faculty_form_unapproved	12	invoice_id -> invoices.id	4 Approved, 8 Pending; 1 unknown invoice (84931); 1 duplicate
3_booking_requests	4	faculty_form_id, invoice_id	Completed, Pending_Batch_Allocation, Failed_Batch_Full, Awaiting_Parent_Batch_Selection
4_class_schedule_bookings	2	bookingrequest_id	Both rows belong to booking 5501 (50-min sessions)
reference_active_batches	13	student_id; batch id	Fee, batch days/time, seats (active vs max 6)

Method

- **Exact identifiers only** link records (invoice id, student id, user id, booking id, batch id). Names are masked and never used; phone similarity is never identity.
- **Deterministic rules** decide payment, duplicates, matching, capacity and eligibility. LLMs are limited to wording and remark classification and cannot change a decision.
- **Every statistic is computed** from the files by `src/metrics.py`; the dashboard and this report read the same outputs. Measured baseline, proposed target, simulation and assumption are labelled separately.
- **Ageing** uses the latest timestamp in the data (2026-11-03 18:00) as 'now' for reproducibility. A single timezone is assumed; the files carry none.
- **Limits.** 13 invoices; masked identifiers (equal masks could hide different numbers); reference table is a student-centred view, not a full batch catalogue; dates span Sunday-Tuesday only, so weekend and holiday delay cannot be measured.

4. Root-cause analysis and actual findings

Twelve findings were observed in the data. Root causes are **hypotheses** to validate with Ops and teachers: the data shows symptoms, not intent (`results/root_cause_analysis.csv`).

#	Finding (evidence)	Likely root cause (hypothesis)
F1	Clean 84903 waits ~33 h for approval	Approval is manual for every row, including clean ones
F2	Paid, no form: 84906 (26 h), 84910 (6.7 h)	Teacher step has no timer, reminder or owner
F3	Form 3313 cites 84931; paid 84913 has no form	Invoice id keyed by hand (digits transposed)
F4	Form 3311 on Failed invoice 84911 (UPI screenshot)	Teacher submits on parent's claim; payment not checked first
F5	84909: form says 10,500 paid; gateway shows 5,250	Instalments flow through the full-payment lane
F6	84907: 20% discount (1,900) offered by teacher, no approver	No discount-code or approval field in any table
F7	84904: payer mobile/email differ from registered	No payer verification step
F8	3308/3309 identical forms about 5 min apart	No idempotency on form submit (browser froze)
F9	Approved 3305 (batch 6/6) and 3312 (no target batch) cannot be booked	Approval happens before capacity/target checks
F10	5502 validated but unscheduled for 49 h	No SLA or owner on approval -> scheduling hand-off
F11	84905 'batch full' although the student already belongs to batch 9205	Capacity check may treat a renewal as a new seat (assumption A1, to validate)
F12	5501: 24 sessions purchased, 2 schedule rows	Rolling scheduling or an incomplete hand-off - unknown

Observed vs theoretical. Not observed in this sample (covered by synthetic tests, theoretical for the company): duplicate invoices, student/teacher/course mismatches, inactive batches, session-level conflicts, multiple children per parent, weekend delay. No timestamp anomalies were found; one initially suspected case was a misreading and the engine correctly raised nothing.

5. Operational leakage audit

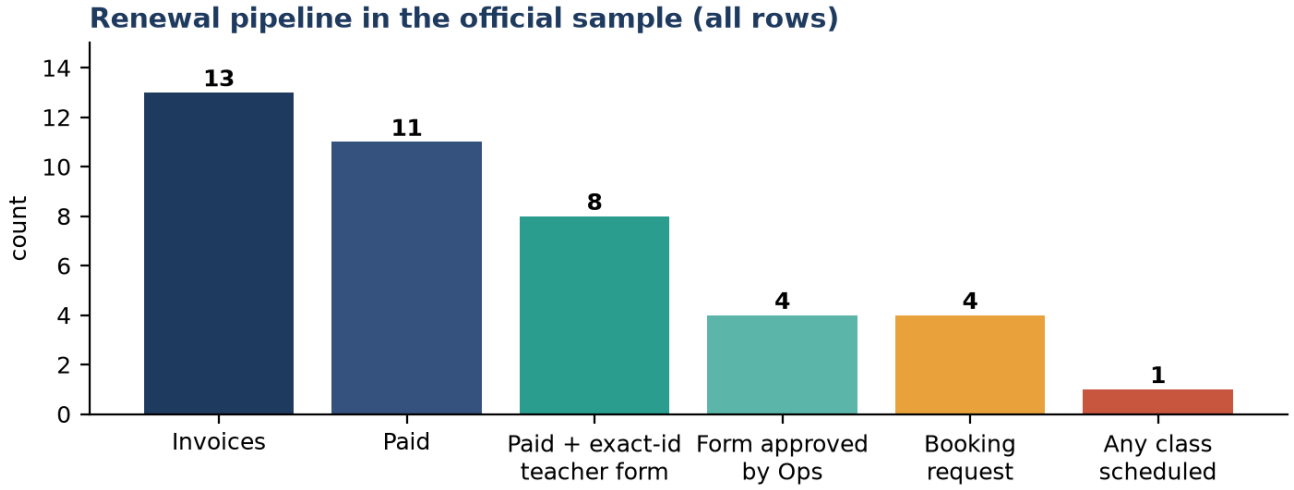


Figure 1 - Only 1 of 11 paid invoices has a scheduled class. Source: files 1-4, all rows.

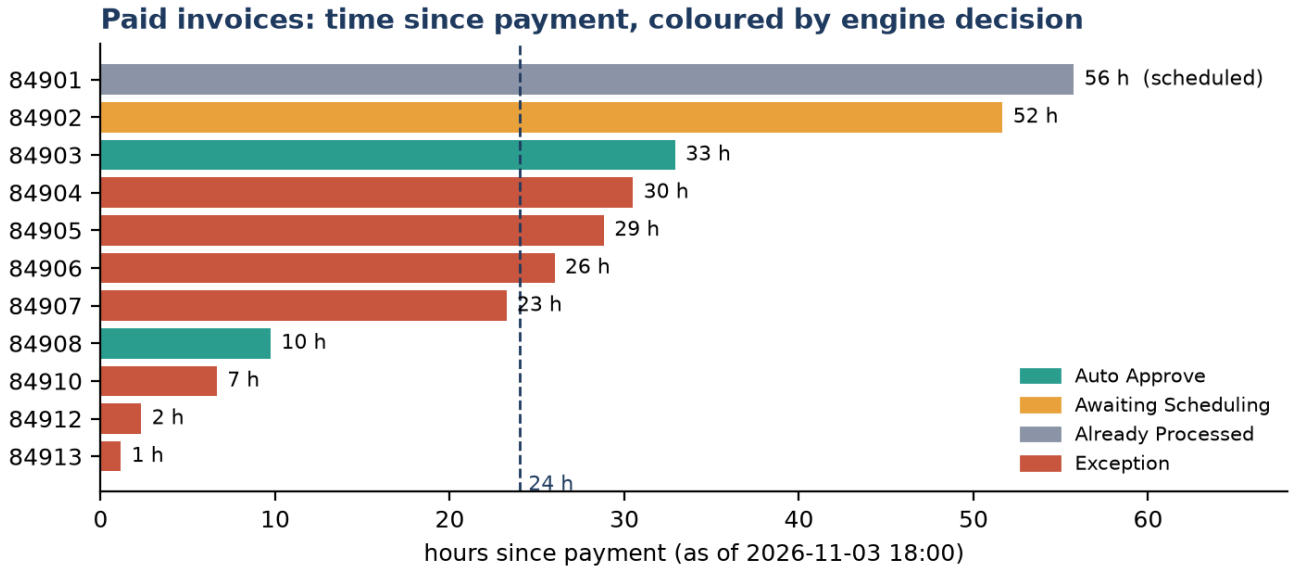


Figure 2 - Five paid invoices (84902-84906) are over 24 h old with no scheduled class. 84903 and 84908 are clean and wait only for approval.

This is a snapshot, not proof of permanent loss: some renewals may be mid-process. It does show where time accumulates: before approval (84903), between approval and scheduling (84902) and in cases that no one owns (84906).

6. KPI and SLA framework

Definitions, numerators, denominators, sources and limitations for every KPI are in `docs/kpi_sla_framework.md` and `results/kpi_table.csv`. Selected measured values:

KPI	Num / Den	Value	Measurable
Paid invoices	11 / 13	84.6%	Yes (n small)
Orphaned paid invoices	3 / 11	27.3%	Yes (n small)
Faculty form submission rate (paid invoices)	8 / 11	72.7%	Yes (n small)
Faculty form mismatch rate	2 / 10	20.0%	Yes (n small)
Duplicate faculty form rate	1 / 13	7.7%	Yes (n small)
Booking requests with no schedule rows	3 / 4	75.0%	Yes (n small)
Payment amount mismatch rate	2 / 13	15.4%	Yes (n small)
Batch capacity issues	1 / 13	7.7%	Yes (n small)
Auto-approval eligibility rate	2 / 13	15.4%	Yes (n small)
Safe-to-schedule rate	4 / 13	30.8%	Yes (n small)
Exception rate	9 / 13	69.2%	Yes (n small)

Measured stage timings (hours)

Stage	n	Median	Max	Reading
Payment -> teacher form created	10	0.45	0.83	Teachers who submit are quick
Form created -> Ops approval	4	1.54	2.5	Approval is the slow human step (approved forms only)
Approval -> booking created	4	0.0	0.0	Booking is created at approval
Booking -> first schedule row	1	0.5	0.5	One observation; 3 bookings have none

Proposed SLAs (targets, not company commitments)

SLA	Target	Measurable today?
Payment detection	5 min from capture	No - no webhook timestamps
Clean renewal reconciliation	5 min after complete data	Not yet - prototype runs in batch
Teacher form reminder / escalation	2 h / 12 h after payment	Yes - 84906 is at 26 h (breach)
P1 payment exception	Ops response within 30 min (operating hours)	No - no exception timestamps yet
Scheduling completion	4 business hours after eligibility	Partly - 5502 at 49 clock hours

Validate with the Operations Head: operating hours and holiday calendar, timezone, and who owns each target.

7. Proposed zero-touch architecture

Principle: deterministic code decides money and eligibility; AI drafts and classifies; humans own exceptions.
Re-engineering moves: gateway as the source of truth; invoice-first renewal (teacher form optional for same-batch); capacity check only for net-new seats; invoice picker instead of free text; discount codes with approver; timers instead of memory.

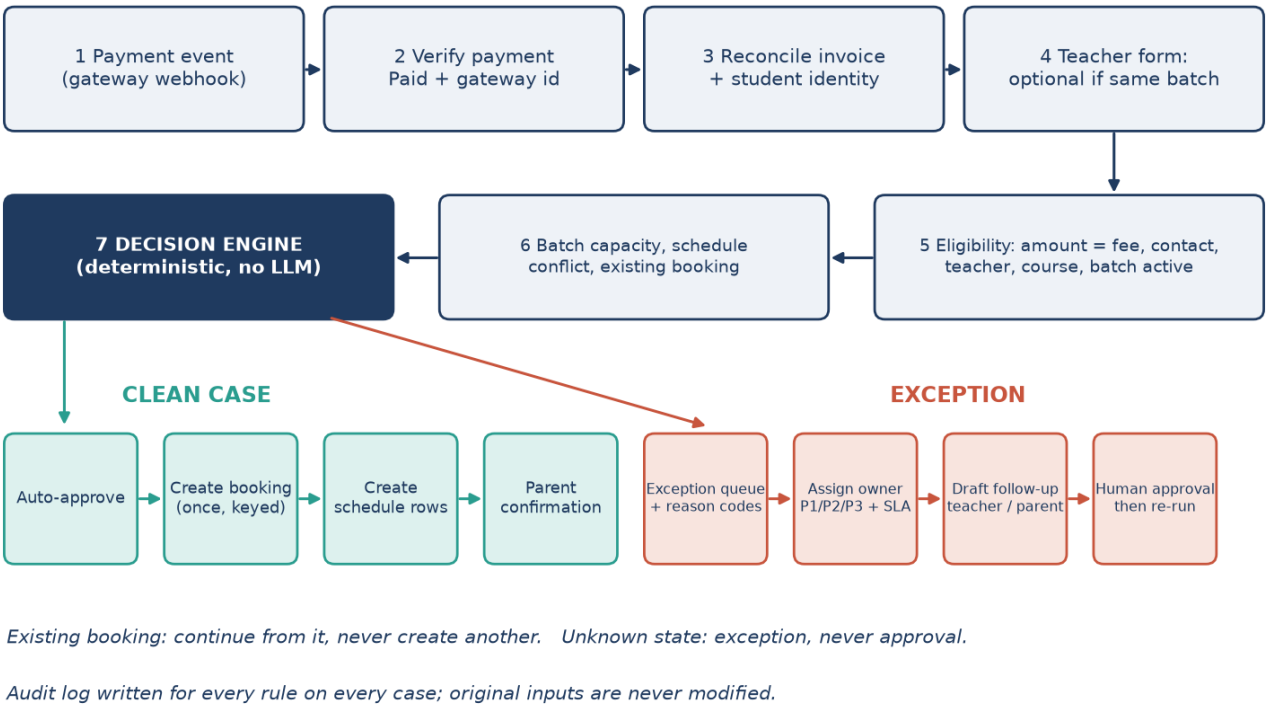


Figure 3 - Target workflow. Clean cases flow to a once-only booking; everything else is owned and re-evaluated after the data is fixed.

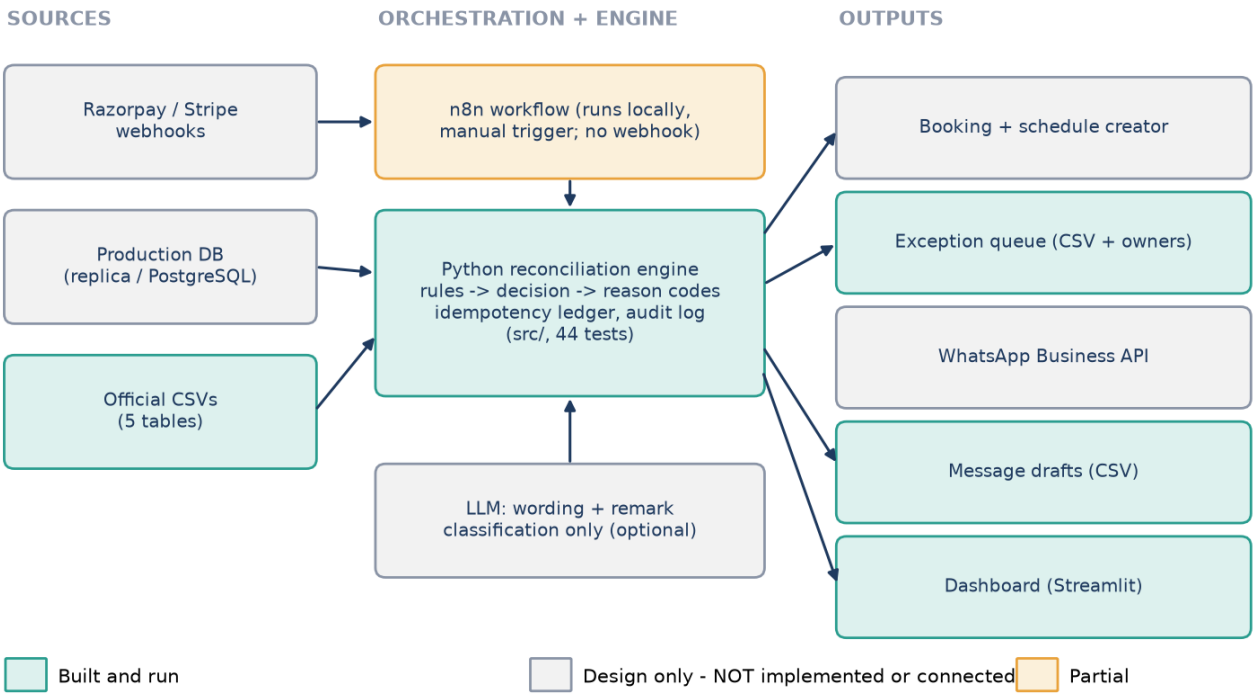


Figure 4 - Components. Green = built and run; amber = partial (the n8n flow runs locally with a manual trigger only, no webhook or real notifications); grey = design only, not implemented or connected.

Questioning the steps before automating them

Step	Why it exists (hypothesis)	Proposal
Teacher renewal form	Captures intent and slot; doubles as a payment claim	Optional for same-batch; mandatory for batch change
Manual payment verification	Teachers submit before payment is confirmed (84911)	Replace with gateway status check
Ops approval	Catches typos, amounts, duplicates	Auto for clean cases; human for the exception classes
Parent / Ops batch pick	Needed when changing batch	Skip for same-batch renewals

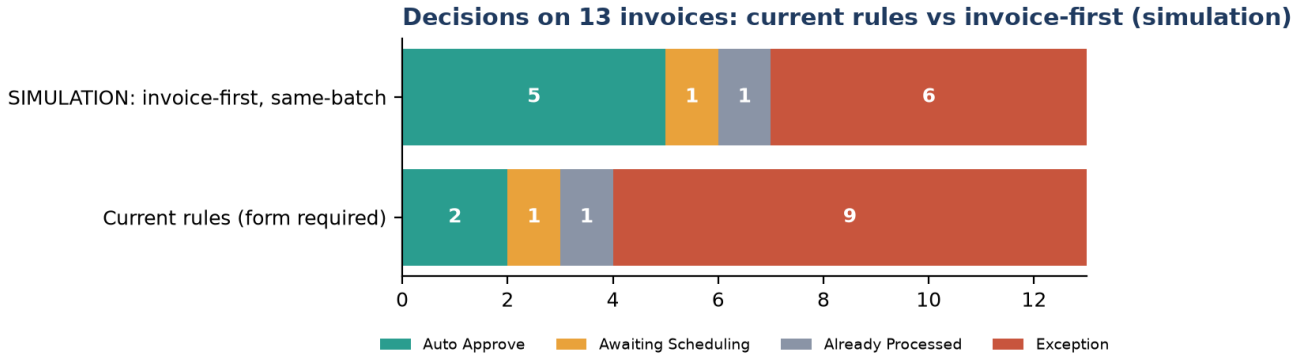


Figure 5 - SIMULATION on the same 13 invoices (not a production result): without the form requirement, auto-approvals rise from 2 to 5 (84906, 84910, 84913 join).

8. Exception taxonomy and renewal decision matrix

A renewal is auto-approved only if **no** blocking reason code is raised. Owners, priorities and SLAs are proposed. Full matrix with detection logic, channel and next action: `docs/decision_matrix.md`.

Scenario	Auto?	Reason code	Owner / Pri	Observed
Same batch, same teacher	Yes	-	monitor / P3	84903, 84908
Same batch, different teacher	No	TEACHER_ID_MISMATCH	Ops Executive / P2	Not observed
Different-batch renewal	No	MISSING_REQUIRED_FIELD + MANUAL_REVIEW	Scheduling Ops / P2	84912
Parent identity mismatch	No	PARENT_CONTACT_MISMATCH	Parent Success / P2	84904
Student identity mismatch	No	STUDENT_ID_MISMATCH	Ops Executive / P1	Not observed
Payment mismatch	No	AMOUNT_MISMATCH / DISCOUNT_REQUIRES_APPROVAL	Finance Ops P1 / Ops Mgr P2	84909, 84907
Partial payment	No	PARTIAL_PAYMENT	Finance Ops / P2	84909
Duplicate invoice	No	DUPLICATE_INVOICE	Finance Ops / P1	Not observed
Missing teacher form	No (today)	MISSING_FACULTY_FORM (+ORPHANED >= 12 h)	Teacher -> Ops Exec / P2	84906, 84910, 84913
Full batch	No	BATCH_FULL	Scheduling Ops / P1	84905
Scheduling conflict	No	SCHEDULING_CONFLICT	Scheduling Ops / P2	Info only: educator 1125 batches overlap Mondays
Existing booking	Never new	BOOKING_ALREADY_EXISTS (info)	Ops Executive / P3	84901, 84902, 84905, 84912

9. Exception handling

Each exception row carries record id, invoice id, student id, all reason codes, explanation, severity, owner, priority, SLA, recommended action, channel and a created timestamp. Closing an exception never edits the engine's decision: the data is fixed at source and the engine re-decides, so the audit trail stays honest.

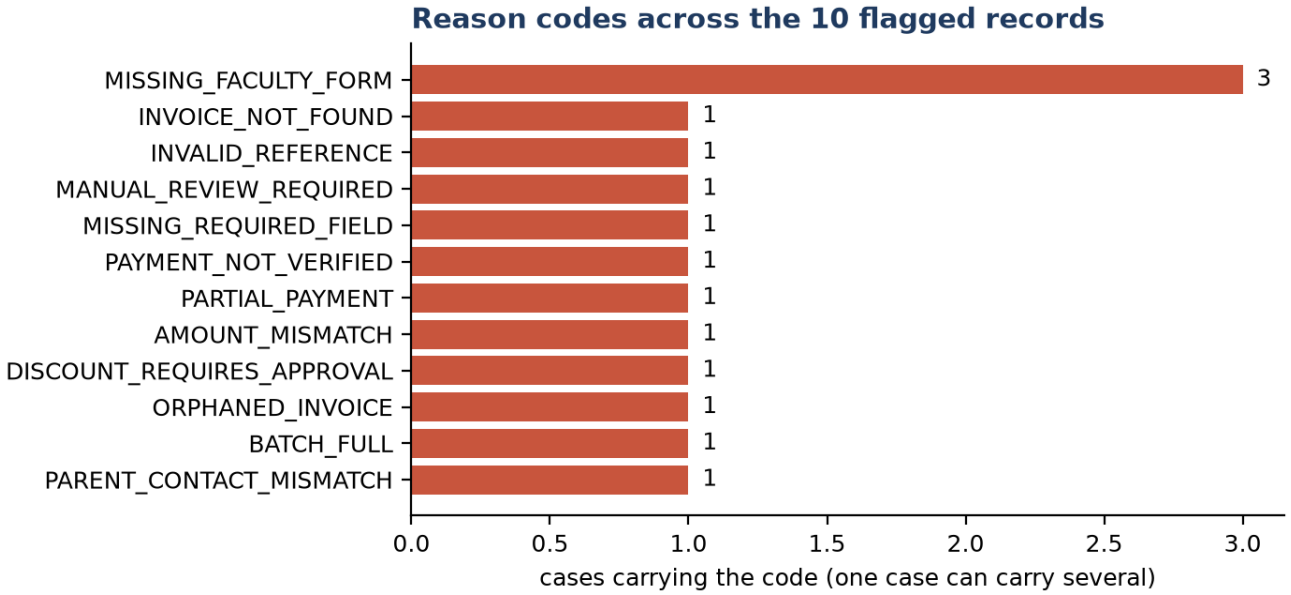


Figure 6 - Reason codes across the 10 flagged records (9 invoices + orphan form 3313).

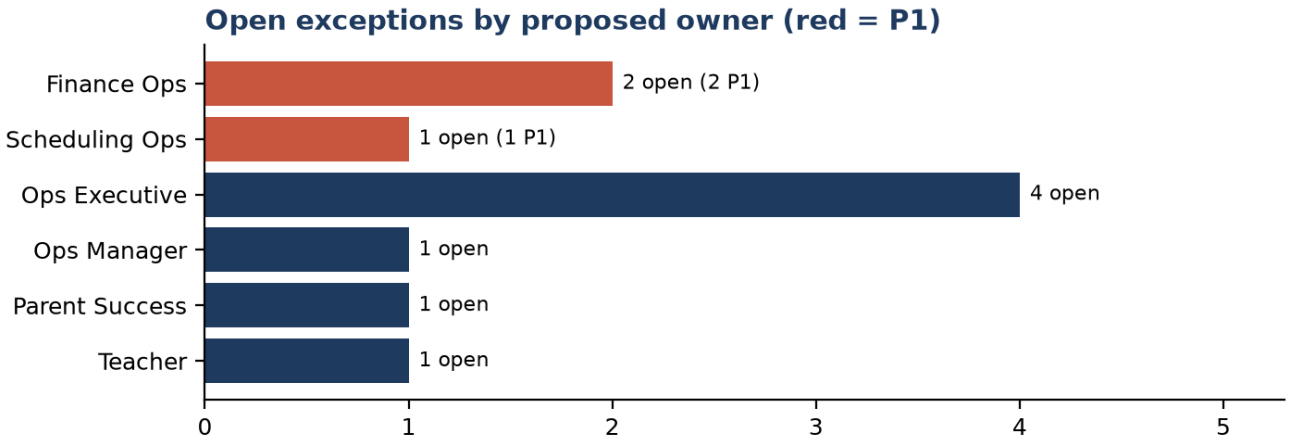


Figure 7 - Proposed ownership of the open queue; red = P1 (84909, 84911, 84905).

Example (84909). Payment status Partially_Paid (5,250 of 10,500) while the teacher entered 10,500. Codes: AMOUNT_MISMATCH + PARTIAL_PAYMENT. Owner Finance Ops, P1. Drafts: teacher amount check, parent instalment reminder, ops escalation. No amounts or gateway ids appear in teacher-facing drafts.

AI's role. Classify free-text remarks (e.g. 'offered 20% retention scholarship discount') and polish drafts. It cannot approve, reject, or change an amount, owner or code. Prompts are in `prompts/`; the LLM path is optional (API key via environment) and was **not tested live**.

10. Prototype implementation

One command runs everything: `python -m src.main --input data/raw --output results`. Dashboard: `streamlit run dashboard/app.py`. Tests: `python -m pytest`.

Module	Responsibility
data_loader / validators	Schema check, NULL handling, phone/email/money/time normalisation (originals kept), input fingerprints
matching_engine / duplicate_detector	Exact-id linkage; probable-match hints (never approve); identical-duplicate form collapse; duplicate invoice detection
eligibility_engine	17 audited steps (16 checks + decision): payment, integrity, duplicates, linkage, identity, teacher/course, contact, amount/discount, required fields, batch, capacity, slot, timeline, booking, conflicts, decision
batch_capacity / scheduling_validator	Seats from reference table (assumption A1 off by default); session and recurring-slot overlaps
exception_handler / message_generator	Owner, priority, SLA, action; deterministic drafts with optional LLM polish; nothing is sent
metrics / report_generator / audit_logger	KPIs with definitions, root-cause table, 206-row audit trail on the sample
main	CLI, idempotency ledger (simulated execution state), all output files

Safety properties demonstrated

- **Idempotency:** first run on the official data creates 3 simulated executions; the second creates 0 and skips 3; outputs are byte-identical (tested).
- **Immutability:** input SHA-256 identical before and after (tested).
- **No silent drops:** every invoice and every orphan form gets a decision; counts reconcile (tested).
- **Fails safe:** unknown state, missing field or fuzzy match produces an exception, never an approval.
- **Tests:** 44 passed, covering the 20 required scenarios plus immutability, idempotency, ownership, messages, metrics, reproducibility and schema errors. Not tested: live LLM, browser rendering, any production integration.

11. Actual prototype results

Case	Decision	Reason codes	Owner / priority
84901	Already processed	info: BOOKING_ALREADY_EXISTS, SCHEDULE_ALREADY_EXISTS	-
84902	Ready: schedule existing booking	info: BOOKING_ALREADY_EXISTS	Scheduling Ops / P1 (49 h wait)
84903	Auto-approve	-	-
84904	Exception	PARENT_CONTACT_MISMATCH	Parent Success / P2
84905	Exception	BATCH_FULL	Scheduling Ops / P1
84906	Exception	ORPHANED_INVOICE, MISSING_FACULTY_FORM	Ops Executive / P2
84907	Exception	DISCOUNT_REQUIRES_APPROVAL	Ops Manager / P2
84908	Auto-approve	info: DUPLICATE_FACULTY_FORM	-
84909	Exception	AMOUNT_MISMATCH, PARTIAL_PAYMENT	Finance Ops / P1
84910	Exception	MISSING_FACULTY_FORM	Teacher / P2
84911	Exception	PAYMENT_NOT_VERIFIED	Finance Ops / P1
84912	Exception	MISSING_REQUIRED_FIELD, MANUAL_REVIEW_REQUIRED	Ops Executive / P2
84913	Exception	INVALID_REFERENCE, MISSING_FACULTY_FORM	Ops Executive / P2
FORM-3313	Exception	INVOICE_NOT_FOUND	Ops Executive / P2

Totals: 13 invoices + 1 orphan form processed; 2 auto-approved (15.4%), 4 safe to schedule or done (30.8%), 9 invoice exceptions (69.2%), 23 follow-up drafts, 206 audit rows. The 15% auto-approval rate is deliberately conservative on a hand-picked edge-case sample; do not read it as a production rate.

84905 nuance. Under assumption A1 (the batch count already includes the student) the 'batch full' reason disappears, but the engine still stops: an existing booking failed and needs a human retry. It never silently re-approves.

12. Operations dashboard

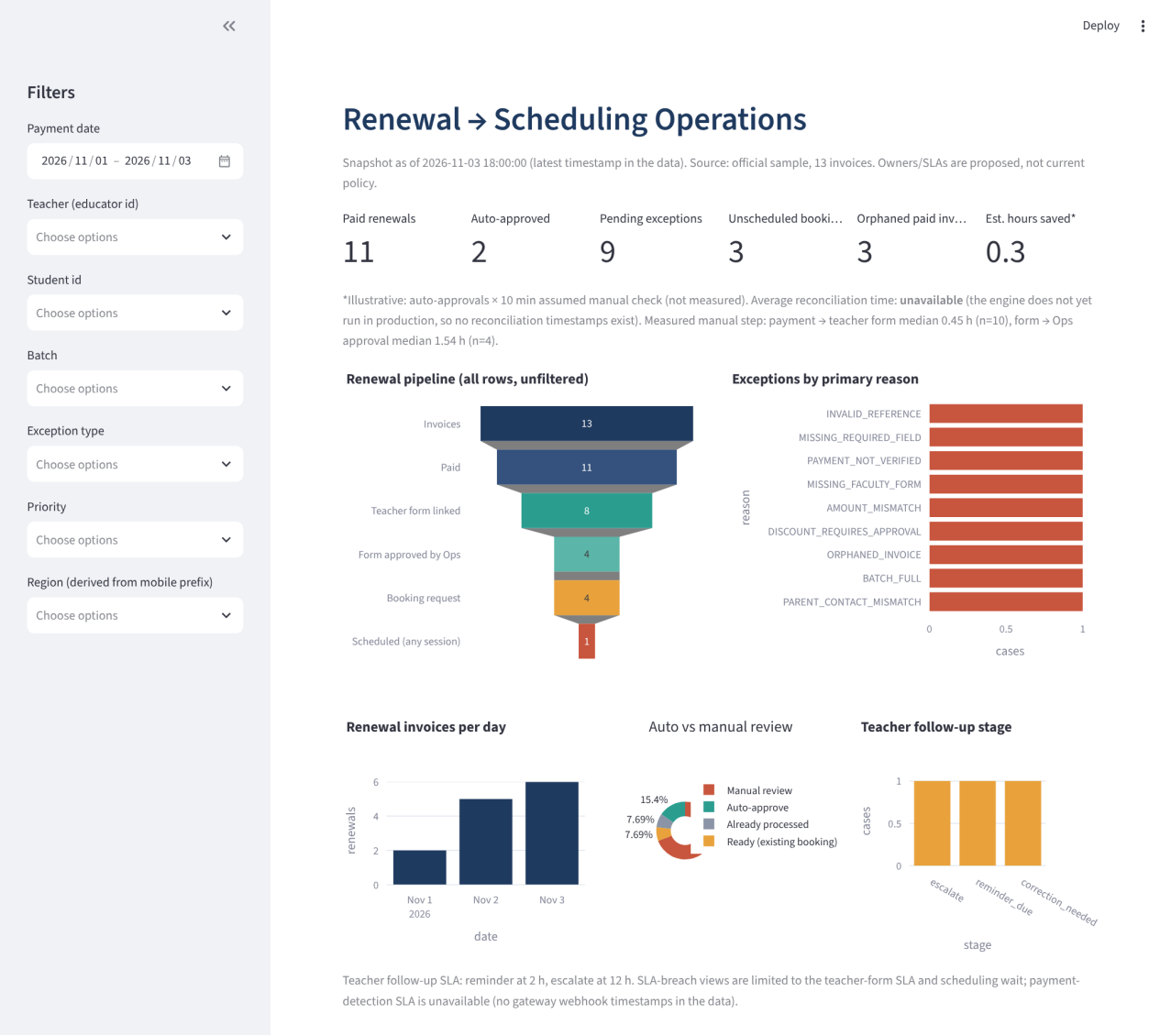


Figure 8 - Screenshot of the running Streamlit dashboard (Chromium via Playwright), top of page. KPIs equal the engine output; filters were tested against independently computed values.

- **KPI cards:** paid renewals, auto-approved, pending exceptions, unscheduled bookings, orphaned invoices, estimated hours saved (illustrative). Average reconciliation time is shown as unavailable: no production reconciliation timestamps exist.
- **Charts:** pipeline funnel, exceptions by reason, daily volume, auto vs manual split, teacher follow-up stage. Payment-detection SLA breaches are unavailable (no webhook timestamps).
- **Tables:** invoices pending over 12 h, forms awaiting submission, bookings awaiting scheduling, payment mismatches, unscheduled students, ownership queue with required action and SLA.
- **Filters:** date, teacher, student, batch, exception type, priority, region (derived from mobile prefix, not a source field).

13. SOP and ownership

The SOP (`docs/operations_sop.md`) covers daily operations, teacher duties, parent communication, exception management, safety and continuity. Owners are **proposed**; the data only shows ops_exec_01 to 03.

Owner (proposed)	Owns	Escalates to
Finance Ops	Payment not verified, amount mismatch, partial payment, duplicate invoice (P1)	Ops Manager if P1 unacknowledged in 30 min
Scheduling Ops	Batch full/inactive, conflicts, batch change, unscheduled bookings	Ops Manager after 4 business h
Ops Executive	Invalid references, orphaned invoices, identity and teacher/course issues	Ops Manager
Parent Success	Parent contact confirmation, batch-change conversations	Ops Executive after 1 business day
Teacher	Submit or correct forms; 2 h reminder, 12 h escalation	Ops Executive
Ops Manager	Discount approval, overrides, policy gaps	Head of Operations

- **Safety:** kill switch on auto-approval; manual override with reason, user and timestamp; append-only audit; rollback by cancelling bookings by idempotency key; role-based data access.
- **Continuity:** during an outage the manual process runs as today and scheduled classes are never touched; on recovery, reconcile the outage window (idempotency prevents duplicates).

14. 30-60-90 day roadmap

Phase	Activities and deliverables	Success metrics	Go / no-go and rollback
Days 1-30 Baseline + shadow	Validate A1, instalment and discount policy; 3-month production extract; time study; engine in shadow mode (no writes); 10-20 teacher pilot; train Ops on queue and dashboard	Engine agrees with Ops outcome on 95% or more of shadow cases; zero false approvals	Go if every disagreement is explained and the rule fixed. Rollback: stop the shadow job (nothing live)
Days 31-60 Controlled auto-approval	50-100 teachers; auto-approve same-batch, full-payment, clean cases only; webhook + n8n + idempotent booking creator; exception notifications (human-approved WhatsApp); weekly false-positive review	Clean renewals scheduled within 4 business h; P1 owner in 30 min; zero duplicate bookings	Go after 2 weeks with zero wrong approvals. Rollback: flip the flag; cancel by idempotency key
Days 61-90 Scale	200+ teachers; form optional for same-batch; batch-change self-service link; Metabase; threshold tuning; savings measured against the 30-day baseline	Auto-approval share vs baseline; payment-to-first-class P50/P90; Ops hours per renewal	Go on KPI targets met 2 weeks plus Head of Ops sign-off. Rollback: cohort back to shadow

Dependencies: read-only DB or exports; webhook signing keys; approved WhatsApp templates; named owners. Change management by stakeholder is in `docs/change_management.md`.

15. Risk management

Risk	Mitigation	Owner
Wrong auto-approval of a payment	Approval needs zero blocking codes; shadow period; kill switch; audit; 0-error go/no-go	Head of Ops
Duplicate booking	Idempotency key + unique constraint; existing booking never recreated (tested)	Engineering
Assumption A1 (seat counted) is wrong	Default off; validate; capacity check for net-new seats	Scheduling Ops
Masked identifiers hide mismatches	Compare full numbers or hashes in production; student_id stays the anchor	Engineering
13-row sample overfits	Treat rates as illustrative until the 3-month extract	Program Manager
Weekend/holiday behaviour untested	Business calendar config; test on real data	Program Manager
LLM drafts hallucinate	Wording only; template fallback; human approval; no new facts	Program Manager
Discount leakage continues	Discount codes with approver id	Ops Manager

Full register with likelihood and impact: `docs/risk_register.md`.

16. Business impact and estimated savings

Model (transparent, assumption-driven): hours saved per month = renewals x auto-approval share x manual minutes avoided / 60. Nothing here is a historical figure.

Input	Value	Basis
Monthly renewals	Unknown (1,000 used only to show scale)	Assumption - obtain from Ops
Auto-approval share, current rules	15.4% (2 of 13)	Measured on sample
Auto-approval share, invoice-first	38.5% (5 of 13)	Simulation on sample
Manual minutes per clean renewal	10	Assumption - run a time study
Hours saved per 1,000 renewals	about 25.6 (current) to about 64.1 (invoice-first)	Output of the model

Beyond hours. Faster time to first class (the sample shows a clean renewal waiting 33 h); fewer approvals of unbookable forms; a control on discounts; visibility of 49-hour stalls. These are valued by the Operations Head, not estimated here.

17. Conclusion and recommendations

- **Start in shadow mode.** Run the engine beside Ops for 2-4 weeks and review every disagreement before any automatic action.
- **Fix the data-entry causes first:** invoice picker, discount codes, form idempotency. They remove whole exception classes.
- **Make the teacher form optional for same-batch renewals** once shadow results confirm the simulation.
- **Validate five assumptions** (seat counting, instalment policy, discount approver, form optionality, owner names) in the first week.
- **Keep AI in its lane:** wording and classification only, behind human approval, until measured error rates justify more.

What is complete and what is not. Complete: data audit, engine, tests, outputs, dashboard, documents, this report. Not complete: a production n8n workflow (the export ran locally only, with a manual trigger and placeholder notifications), payment/WhatsApp/DB integrations, live LLM test, the optional Loom walkthrough.

Appendix A. Reason-code taxonomy (rubric)

Machine-readable source of truth: `src/config.py`. 'Blocks' = prevents auto-approval while open. Informational codes never page anyone.

Code	Blocks	Severity	Owner	Pri	Required action
PAYMENT_NOT_VERIFIED	Yes	Critical	Finance Ops	P1	Check gateway dashboard for the order; confirm captured/failed/pending before any approval.
INVOICE_NOT_FOUND	Yes	High	Ops Executive	P2	Locate the intended invoice (see probable match) and ask the teacher to resubmit with the right invoice.
ORPHANED_INVOICE	Yes	High	Ops Executive	P2	Paid invoice has no form and no probable candidate form; contact teacher/parent to identify the renewal.
MISSING_FACULTY_FORM	Yes	High	Teacher	P2	Teacher submits the renewal form (reminder at 2h, escalate to Ops Executive at 12h).
DUPLICATE_INVOICE	Yes	High	Finance Ops	P1	Confirm which invoice is authoritative; refund or merge the duplicate.
DUPLICATE_FACULTY_FORM	No	Low	Ops Executive	P3	Identical duplicate collapsed to the earliest form; non-identical duplicates need review.
AMOUNT_MISMATCH	Yes	High	Finance Ops	P1	Reconcile invoice net amount vs teacher-entered amount vs standard fee before approving.
DISCOUNT_REQUIRES_APPROVAL	Yes	Medium	Ops Manager	P2	Confirm the discount was authorised by Sales/Ops Head; record the approver.
STUDENT_ID_MISMATCH	Yes	Critical	Ops Executive	P1	Verify student identity with parent/teacher; never approve on name or phone similarity.
PARENT_CONTACT_MISMATCH	Yes	Medium	Parent Success	P2	Confirm with the registered parent number that the paying contact is authorised.
TEACHER_ID_MISMATCH	Yes	High	Ops Executive	P2	Confirm the educator for the batch; correct the form or the batch record.
COURSE_MISMATCH	Yes	High	Ops Executive	P2	Confirm the course/level being renewed.
BATCH_INACTIVE	Yes	High	Scheduling Ops	P1	Find an equivalent active batch and agree it with the parent.
BATCH_FULL	Yes	High	Scheduling Ops	P1	Check whether the student's seat is already counted (same-batch renewal); otherwise offer alternatives.
SCHEDULING_CONFLICT	Yes	High	Scheduling Ops	P2	Resolve teacher/student overlap before creating schedule rows.
BOOKING_ALREADY_EXISTS	No	Low	Ops Executive	P3	Do not create a new booking; continue from the existing one.
SCHEDULE_ALREADY_EXISTS	No	Low	Ops Executive	P3	Do not recreate schedule rows.
PARTIAL_PAYMENT	Yes	High	Finance Ops	P2	Hold scheduling per policy; send instalment-2 reminder; confirm instalment policy with Ops Head.
MISSING_REQUIRED_FIELD	Yes	Medium	Ops Executive	P2	Obtain the missing field (e.g. target batch) from teacher/parent.
INVALID_REFERENCE	Yes	High	Ops Executive	P2	Fix the dangling/typo reference with the teacher; use the probable match as a hint only.
TIMESTAMP_ANOMALY	Yes	Medium	Ops Executive	P2	Form was created before its invoice existed; verify the form really belongs to this invoice.
MANUAL_REVIEW_REQUIRED	Yes	Medium	Scheduling Ops	P2	Human decision needed (e.g. batch-change renewal).

Sources and assumptions

- **Source:** official repository github.com/bambinoslive/AI-PM-hiring-take-home-assignments (ops/data, ops/README). Reproduce with `python -m src.main`.
- **A1** active_students_count may already include the renewing student (default: off). **A2** all timestamps share one timezone. **A3** 50-minute sessions (from schedule rows). **A4** 10 manual minutes per clean renewal. **A5** owner roles, priorities and SLAs are proposals. **A6** currency INR on every invoice, including Stripe +971/+1, taken at face value.
- **Repository note:** the official README restricts these assets to shortlisted candidates and gives no sharing terms, so the raw CSVs are excluded from git history (see .gitignore). Derived outputs contain masked values from the same data: share the repository privately or confirm with Bambinos first.